ANALYSIS REPORT

Polished RCIM Model-Bank Reproduction Campaign Results Report

Polished RCIM Model-Bank Reproduction Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

The `polished_dataset_rcim_model_bank_reproduction_2026_06_22` campaign completed both direction-specific `RCIM Model-Bank Reproduction` runs on `polished_dataset`.

The accepted surfaces are

Surface	Winner	Mean MAPE %	Mean MAE	Mean RMSE	Export
`forward`	`ERT / ExtraTreesRegressor`	11.939192	0.062217	0.149061	`190` Python, `190` ONNX
`backward`	`ERT / ExtraTreesRegressor`	18.399598	0.043815	0.110706	`190` Python, `190` ONNX

Both surfaces used `data\polished_dataset`, split `969` direction-specific curves into `678` train, `194` validation, and `97` test rows, and exported without ONNX failures.

This is a normal campaign closeout. It accepts the polished RCIM model-bank reproduction artifacts and updates program status, but it does not run or replace the official `TE Curve Verification Pipeline`. Curve-first official promotion remains a separate operator-approved workflow.

Campaign Scope

The campaign reran the paper-faithful RCIM model-bank reproduction workflow on the measured `polished_dataset` schema. The source CSV measurements are the polished point columns:

Column	Role
`theta`	motor position measured in degrees
`theta_dot`	motor velocity derived from position
`tau_load`	applied load in Nm
`T`	oil temperature
`theta_TE`	measured transmission error target

For the RCIM exact-paper workflow, each curve is transformed into the paper-compatible feature schema:

RCIM Feature	Source Meaning
`rpm`	measured curve speed
`deg`	oil temperature condition
`tor`	measured load / torque condition

The target surface is the selected harmonic amplitude/phase bank for harmonics `0`, `1`, `3`, `39`, `40`, `78`, `81`, `156`, `162`, and `240`.

Execution Summary

Surface	Run Instance	Completion Evidence
`forward`	`2026-06-22-23-42-04__rcim_model_bank_reproduction_polished_dataset_fw_polished_dataset_campaign_validation`	completed `2026-06-23 18:47:42`; validation summary, best-parameter summary, model bundle, Python export, and ONNX export present
`backward`	`2026-06-25-15-19-40__rcim_model_bank_reproduction_polished_dataset_bw_polished_dataset_campaign_validation`	completed `2026-06-26 09:33:39`; validation summary, best-parameter summary, model bundle, Python export, and ONNX export present

The backward run was resumed with

`powershell .\scripts\campaigns\cross_wave\run_polished_dataset_rcim_model_bank_reproduction_campaign.ps1 -Surface bw`

The final backward log ends with

`text Polished-dataset RCIM Model-Bank Reproduction validation complete`

Directional Results

Forward

- dataset root: `data\polished_dataset`
- direction label: `forward`
- rows: `969`
- split: `678` train / `194` validation / `97` test
- winner: `ERT / ExtraTreesRegressor`
- mean component MAPE: `11.939192%`
- mean component MAE: `0.062217`
- mean component RMSE: `0.149061`
- Python exports: `190`
- ONNX exports: `190`
- ONNX failed exports: `0`

Backward

- dataset root: `data\polished_dataset`
- direction label: `backward`
- rows: `969`
- split: `678` train / `194` validation / `97` test
- winner: `ERT / ExtraTreesRegressor`
- mean component MAPE: `18.399598%`
- mean component MAE: `0.043815`
- mean component RMSE: `0.110706`
- Python exports: `190`
- ONNX exports: `190`
- ONNX failed exports: `0`

Artifact Paths

Artifact	Path
campaign leaderboard	`output\training_campaigns\cross_wave\polished_dataset\rcim_model_bank_reproduction\polished_dataset_rcim_model_bank_reproduction_2026_06_22\campaign_leaderboard.yaml`
campaign best-run YAML	`output\training_campaigns\cross_wave\polished_dataset\rcim_model_bank_reproduction\polished_dataset_rcim_model_bank_reproduction_2026_06_22\campaign_best_run.yaml`
campaign best-run Markdown	`output\training_campaigns\cross_wave\polished_dataset\rcim_model_bank_reproduction\polished_dataset_rcim_model_bank_reproduction_2026_06_22\campaign_best_run.md`
forward validation summary	`output\validation_checks\rcim_model_bank_reproduction\2026-06-22-23-42-04__rcim_model_bank_reproduction_polished_dataset_fw_polished_dataset_campaign_validation\validation_summary.yaml`
backward validation summary	`output\validation_checks\rcim_model_bank_reproduction\2026-06-25-15-19-40__rcim_model_bank_reproduction_polished_dataset_bw_polished_dataset_campaign_validation\validation_summary.yaml`
forward validation report	`doc\reports\analysis\validation_checks\2026-06-23-18-47-42_rcim_6c4cd030_rcim_model_bank_rep_a049be30_polished_dataset_rcim_model_bank_reproduction_report.md`
backward validation report	`doc\reports\analysis\validation_checks\2026-06-26-09-33-39_rcim_6c4cd030_rcim_mod_4166ebbf_polished_dataset_rcim_model_bank_reproduction_report.md`

GitHub Artifact Boundary

The generated model bundles are intentionally not Git-tracked:

Surface	Bundle Size	Reason
`forward`	`151.82 MB`	exceeds GitHub `100 MB` single-file limit
`backward`	`145.96 MB`	exceeds GitHub `100 MB` single-file limit

The local generated export folders are also left as local artifacts for this closeout. The repository records the summaries, reports, registry updates, and campaign acceptance metadata.

Registry And Status Effects

The program best-parameter registry was updated with the backward `polished_dataset` RCIM model-bank entries on `2026-06-26T09:33:26`.

This closeout does not change the scalar neural program winner `te_periodic_gru_sequence_remote_global`, and it does not replace the accepted direction-parallel `TE Curve Verification Pipeline` leaders:

- forward curve-verified leader: `rcim_retuned_GBM19_Fw`;
- backward curve-verified leader: `periodic_gru_sequence_Bw`;
- global neural curve-verified leader: `periodic_gru_sequence_global`.

Acceptance Decision

The campaign is accepted because

- both directional runs completed;
- both runs used `data\polished_dataset`;
- both runs produced validation summaries and best-parameter summaries;
- both runs exported `190` Python artifacts and `190` ONNX artifacts;
- no ONNX export failures were reported;
- campaign-level winner artifacts now expose the accepted surfaces.

Follow-Up

Recommended next steps

1. keep the official `TE Curve Verification Pipeline` refresh separate from this closeout;
2. launch the already prepared full-wave polished retraining campaign only after this closeout is committed;
3. if RCIM polished candidates should enter official curve-first comparison, prepare a separate operator-approved verification launcher.